

Supervision Construction

Dense targets are derived after spatial augmentation so their geometry and morphological widths match the tensors passed to the network. All radii below refer to the 384 x 384 training grid.

Let $T(x) \in \{0, \dots, 8\}$ be the semantic target, $B(x) \in \{0, 1\}$ the Fish Region, and $P(x) \in \{0, \dots, 4\}$ the Part Map. The symptom foreground is

$$M(x) = \mathbf{1}[T(x) > 0]$$

Part-structure information S

The binary Fish Region B supervises Fish Region prediction over the full image. The reviewed Part Map P supervises the five-class Part Map inside the fish.

Head-to-Tail Direction is generated as a scalar coordinate in $[0, 1]$ inside the Fish Region. When mouth and caudal-fin pixels are present, their centroids fix the head and tail endpoints. Otherwise, the principal axis of the Fish Region supplies the direction and an available endpoint fixes its sign. Outside-fish values are set to -1 .

Zone Map contains body, mouth, fin tip, fin middle, fin base, caudal-fin tip, caudal-fin middle, and caudal-fin base. For each appendage, the base is the region touching a body dilation of radius 3, the tip follows the Fish Region boundary with radius 2 and excludes the base, and the middle is the remaining appendage region.

Fish Region and Part Map provide the weighted Part-structure objective. The direction and zone targets define the corresponding reference maps and route targets; their network maps are formed continuously from the predicted fish geometry and Part Map.

Visual evidence E

Redness, Shape, and Lesion

The three positive Visual evidence targets are soft, class-dependent labels on M :

Symptom class	Redness	Shape	Lesion
Body lesion	0.10	0.00	1.00
Mouth ulcer	0.35	0.00	0.95
Fin deformity	0.00	1.00	0.75
Fin necrosis	0.15	0.20	1.00
Fin base necrosis	0.15	0.20	1.00
Caudal deformity	0.00	1.00	0.75
Caudal necrosis	0.15	0.20	1.00
Caudal base necrosis	0.15	0.20	1.00

Every annotated symptom receives a Lesion target of at least 0.75. The valid mask for these losses is M ; unannotated within-fish pixels are not used as negative Redness, Shape, or Lesion examples.

Unaffected Surface

Let $\mathcal{D}_r$ and $\mathcal{E}_r$ denote dilation and erosion with an elliptical radius- r kernel. A Part Map boundary ∂P is the union of the radius-1 morphological gradients of the foreground parts. The positive Unaffected Surface target is

$$T_U = \mathcal{E}_5(B) \setminus (\mathcal{D}_7(M) \cup \mathcal{D}_3(\partial P))$$

This excludes the fish edge, a seven-pixel symptom margin, and a three-pixel part-transition margin. Unaffected Surface is supervised as 1 on T_U and 0 on M , with valid region $T_U \cup M$.

Valid regions

Quantity	Valid pixels
Fish Region	full image
Part Map	B
Redness, Shape, Lesion	M
Unaffected Surface	$M \cup T_U$
Class responses R_0 , N , and C	M
Candidate foreground and suppression terms	$M \cup T_U$
Route assignment, Spatial Support, route interpretation, and Z	M ; T_U supplies the explicit none/negative region where used
Native and Auxiliary semantic outputs	$M \cup \overline{B}$
Final-refinement boundary terms	Q_{boundary}

Restricting semantic supervision to annotated symptoms and image background prevents an unannotated within-fish pixel from becoming an automatic symptom-free semantic label.

Route assignment and Spatial Support

Symptom class	Route target	Spatial Support target
Body lesion	body	body
Mouth ulcer	mouth	mouth
Fin deformity	fin	fin tip
Fin necrosis	fin	fin middle
Fin base necrosis	fin	fin base
Caudal deformity	caudal fin	caudal-fin tip

Symptom class	Route target	Spatial Support target
Caudal necrosis	caudal fin	caudal-fin middle
Caudal base necrosis	caudal fin	caudal-fin base

Route assignment also receives Part Map consistency supervision over fish-part pixels. Route-specific class heads use the annotated class in their allowed route, with Unaffected Surface contributing a weak none-class valid region.

Final refinement

The boundary band is

$$Q_{\text{boundary}} = \mathcal{D}_5(M) \setminus \mathcal{E}_5(M)$$

A signed distance transform of M , clipped to 16 pixels, supervises boundary placement in this band. Foreground Dice, curvature, edge-weighted total variation, and bilateral smoothness regularize the refined output.

Target construction is implemented in [data/dataset.py](#), and the masks are consumed by [losses/objective.py](#).